KRISHNA VERMA

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Professional Summary

Electronics and Communication Engineering undergraduate (B.Tech, PSIT Kanpur) focused on digital design, VLSI, and hardware systems. Building an 8-bit CPU from scratch in Verilog, starting from Boolean logic and gate-level design. Hands-on experience across the hardware stack: 4-layer PCB design, embedded systems (ESP32, Arduino, Raspberry Pi), UAV and RC builds, and electronics prototyping. Completed VLSI and Embedded training from NIELIT (MeitY). Detail-oriented, quick to learn, and a dependable team player seeking to start a career in semiconductor engineering.

Target Roles: Physical Design / APR Engineer | Process Engineer | Equipment Engineer | Process Integration Engineer

Education

B.Tech, Electronics and Communication Engineering 2023 – Present (Expected 2027)
PSIT, Kanpur, Uttar Pradesh, India | CGPA: 7.5/10

Senior Secondary (XII), Science (Mathematics) 2022 – 2023
Guru Nanak Public School, Kanpur, Uttar Pradesh, India

Relevant Coursework

Digital Electronics | Analog Electronics | Microcontrollers | Signals and Systems | Communication Systems

Technical Skills

Digital Design & VLSI: Verilog HDL, digital electronics, Boolean logic and logic minimization, combinational and sequential circuits, finite state machines, computer architecture (ISA, ALU, datapath, control unit), testbenches and waveform simulation, basic VLSI concepts
Programming: C++, Python, Linux (basic), MATLAB
PCB Design & EDA: KiCad, EasyEDA, schematic capture, multi-layer PCB layout, design rule checks (DRC), Gerber generation
Embedded & Hardware: Arduino, ESP32, Raspberry Pi, microcontrollers, IoT systems, analog and digital circuit design, datasheet analysis
Tools: Fusion 360, VS Code, Arduino IDE, Git (learning)

Projects

8-Bit CPU Design from Scratch (In Progress) 2026
Verilog, Digital Logic

- Designing a complete 8-bit CPU from the ground up in Verilog, starting from logic gates and Boolean expressions, with a custom instruction set architecture (ISA).
- Deriving truth tables and simplifying logic expressions (Boolean algebra, K-maps) to build core blocks: ALU, registers, program counter, instruction decoder, and control unit.
- Writing modular, reusable Verilog for each block and verifying it with testbenches and waveform analysis before integrating the full datapath.

ESP32-C3 4-Layer Multi-Sensor PCB Design 2026
KiCad, EasyEDA

- Designed a custom 4-layer PCB around the ESP32-C3 integrating multiple sensor interfaces: schematic capture, multilayer routing, design rule compliance, and Gerber generation for fabrication.
- Optimized layout for signal integrity, compact form factor, and manufacturability.

Autonomous Surveillance Drone (FPV-Enabled) 2025
Embedded Systems, UAV

- Built an FPV autonomous UAV integrating GPS, flight controller, ESCs, brushless DC motors, and wireless modules.
- Performed calibration, iterative flight testing, and parameter tuning to improve stability and robustness.

RC Aircraft: F-22 Raptor Model 2025
Aerodynamics, Embedded Electronics

- Designed and fabricated a functional RC fixed-wing aircraft with embedded electronic control and RF-based remote operation.
- Ran multiple test flights to refine aerodynamic stability, weight distribution, and control response.

Professional Experience

Jr Project Associate, IoT & Embedded Systems Jul 2025 – Feb 2026
Tackle Studios

- Worked with a multidisciplinary engineering team on embedded systems, electronics, and hardware development for product-oriented research.
- Performed datasheet analysis and comparative evaluation of electronic components to support design decisions.
- Prepared structured technical documentation for ongoing engineering projects.

Certifications

VLSI for Beginners, NIELIT Calicut (MeitY) | Embedded for Beginners, NIELIT Calicut (MeitY) | Generative AI, Oracle Cloud Infrastructure | Artificial Intelligence Fundamentals, IBM | AI and Cybersecurity Awareness, TCS iON | Python Programming for Beginners, Udemy

Achievements

PR1 Finalist, NIDAR Drone Competition | Participant, Smart India Hackathon (SIH) Internal Round 2025

Strengths & Languages

Strengths: Structured problem solving | Debugging and root-cause analysis (hardware and code) | Technical research and documentation | Team collaboration | Learning quickly in fast-paced environments
Languages: English (Professional) | Hindi (Native)